SCI Peer-Review Defense & Rebuttal Playbook (15 Questions)

Manuscript: Governing Human–AI Collaborative Project Workflows in Multi-Platform Digital Delivery: A Design Science Study of Auditable Handoffs and Idempotent Release Operations

Author: HuaiCheng Sheng (Agentic Work Lab, Hangzhou Chengxin Holding Group Co., Ltd. | ORCID: `0009-0003-2900-6377`)

This playbook anticipates the 15 toughest questions from SCI Q1 reviewers across Project Management Theory (Reviewer #1), Distributed Systems & State Machines (Reviewer #2), and Empirical Statistics (Reviewer #3), providing ready-to-copy English responses and Chinese strategic summaries.

Q1 [Reviewer #1 (Project Management & DSR Theory)]
EN: How does your four-layer architecture advance Stage-Gate project management theory rather than merely implementing standard DevOps CI/CD scripts?
CN: 你们的四层架构是如何在理论上推进项目管理“阶段门控（Stage-Gate）”理论的，而不是仅仅写了几个普通的 DevOps 自动化脚本?

Standard SCI Rebuttal (English): Classical Stage-Gate theory (Cooper, 1990; Bianchi et al., 2022) assumes human-to-human review meetings where actors share semantic context, whereas classic CI/CD pipelines (Humble & Farley, 2010) assume deterministic compilers and idempotent deployment targets. Multi-agent AI workflows violate both assumptions: foundation models suffer from stochastic context drift across asynchronous sessions, and external digital platforms exhibit non-idempotent, opaque state transitions (e.g., 92.32% post-submission confirmation waits). Our framework advances Stage-Gate theory by formalizing 'Computational Stage-Gates' through three enforceable invariants: (1) pure temporal-checklist scheduling predicates (Algorithm 1, Theorem A.1), (2) cryptographic plan-hash binding between dry-run rehearsal and execution (--confirm CURRENT_PLAN_HASH, Theorem A.3), and (3) epistemic audit gates (Algorithm 3) that prevent agents from conflating raw output volume with verified project outcomes.

核心防御逻辑：传统 Stage-Gate 假设“人与人开会评审”，传统 CI/CD 假设“编译器和部署目标是确定且幂等的”。但多智能体项目同时打破了这两个假设（AI 会上下文漂移，外部社交平台 92.32% 时间卡在不确定等待态）。我们将阶段门控理论升华为“可计算门控不变式（Computational Stage-Gates）”，并在 Appendix A 给出了形式化定理证明（Theorems A.1–A.3）。

Q2 [Reviewer #1 (Project Management & DSR Theory)]
EN: Why is a 'Sole-Operator' human-in-the-loop (HITL) topology necessary when frameworks like ChatDev and MetaGPT advocate fully autonomous multi-agent teams?
CN: 既然 ChatDev 和 MetaGPT 等框架都在鼓吹“全自动无人值守多智能体团队”，为什么你们还要坚持“单人最高授权（Sole-Operator HITL）”?

Standard SCI Rebuttal (English): As demonstrated in our comparative governance analysis (Table 1) and ablation study (Table 11, Ablation A4/A5), fully autonomous conversational handoffs lack legal/operational accountability and fail catastrophically when encountering non-API authentication barriers (CAPTCHAs, QR re-logins) or ambiguous timeout states. Drawing on high-reliability control theory (Parasuraman et al., 2000; Shneiderman, 2020; Raisch & Krakowski, 2021), high automation must be paired with high human control over irreversible external commitments. By partitioning roles so that GPT/Codex, Claude, and DeepSeek handle high-frequency synthesis, UI engineering, and heartbeat monitoring while the human Project Manager retains sole release-token authority, our system achieved 0 unauthorized writes across N=69 checkpoints while bounding human intervention solely to explicit NEEDS_HUMAN boundary states.

核心防御逻辑：全自动多智能体在遇到滑块验证码、Cookie 过期或超时重发时会直接崩溃或闯祸（见表 1 对比与表 11 消融实验 A4/A5）。依据 Shneiderman (2020) 的“高自动化+高人类控制”理论，把高频生成/巡检交给 AI，把不可逆的外部发布锁（Sole Operator）留给人类项目经理，才能实现 69 次运行 0 越权事故。

Q3 [Reviewer #1 (Project Management & DSR Theory)]
EN: How does this study fulfill the six mandatory steps of Peffers et al.'s (2007) Design Science Research Methodology (DSRM) and Gregor & Hevner's (2013) knowledge contribution framework?
CN: 本研究是如何严格满足 Peffers (2007) 的设计科学研究六步法以及 Gregor & Hevner (2013) 的知识贡献框架的?

Standard SCI Rebuttal (English): Section 3.1 and Figure 1 explicitly map all six DSRM activities: (1) Problem Identification (three failure modes in Section 1), (2) Objectives of a Solution (four-layer invariants), (3) Design & Development (Algorithms 1–3, 123 Python modules, 27,095 LOC), (4) Longitudinal Demonstration (two multi-episode production series and standalone releases across 5 platforms), (5) Multi-Method Evaluation (N=69 operational events, N=55 channel records, N=30 SHA-256 media versions, and 2,737 unit tests), and (6) Communication (complete open-source archival and 150-page Supplementary Material). Under Gregor & Hevner's (2013) matrix, our work provides an 'Exaptation / Improvement' contribution that formalizes nascent design theory for agentic project governance.

核心防御逻辑：第 3.1 节完整对应了 DSRM 六步法，兼具软件制品（27,095 行核心代码）与纵向实证评估（69 条运维日志、55 条发布快照、2,737 个测试），符合 Gregor & Hevner (2013) 的设计科学知识贡献标准。

Q4 [Reviewer #2 (Distributed Systems & State-Machine Formalism)]
EN: In Theorem A.2 and Algorithm 2, why do you latch post-submission verification timeouts into an 'UNKNOWN' state rather than implementing automatic exponential-backoff retries?
CN: 在定理 A.2 和算法 2 中，为什么提交后回读超时要锁定为 UNKNOWN 状态，而不是像常规网络程序那样做“指数退避自动重试”?

Standard SCI Rebuttal (English): Automatic exponential-backoff retry is only safe when the target operation is natively idempotent (e.g., HTTP PUT with a server-enforced idempotency key). Across heterogeneous social media platforms (especially browser-automated endpoints like Douyin and Instagram or chunked uploads without universal idempotency tokens), a timeout during the post-submission confirmation window (which consumes 92.32% / 380.551 s of Instagram release duration, Table 5) means the payload has already been transmitted and the server is likely processing it asynchronously. As proven in Theorem A.2 (Supplementary Appendix A) and empirically confirmed on Douyin (Table 4), automatic retry after SUBMITTING violates At-Most-Once publication safety and produces duplicate public posts. Latching into UNKNOWN and restricting recovery strictly to read-only reconciliation (verify_remote_state) guarantees zero duplicate side effects across K >= 1 recovery rounds.

核心防御逻辑：指数退避重试只对幂等接口安全！而在抖音、Instagram 等跨平台发布中，92.32% 的耗时发生在“点击提交后的等待确认期”（见表 5）。此时视频数据早已上传完毕，如果因为超时就自动重传，100% 会造成重复发帖（抖音实测案例见表 4）。锁定为 UNKNOWN 并只允许只读回查，在数学上（定理 A.2）保证了“最多发布一次（At-Most-Once）”。

Q5 [Reviewer #2 (Distributed Systems & State-Machine Formalism)]
EN: What threat model does the cryptographic plan-hash lock (--confirm CURRENT_PLAN_HASH) defend against?

CN: 你们设计的加密计划哈希锁（--confirm CURRENT_PLAN_HASH）具体防御的是什么威胁模型（Threat Model）？

Standard SCI Rebuttal (English): The plan-hash lock defends against Time-of-Check to Time-of-Use (TOCTOU) drift in multi-agent pipelines. Specifically, between the time a human Project Manager reviews a dry-run rehearsal plan and the moment the execution agent invokes the live release command, another asynchronous agent or background script might modify the title, tags, target platform list, or re-render the video file. Because H_plan = SHA256(Canon(M, T, H(f))) binds the canonical metadata M, target platform vector T, and full stream-based SHA-256 media digest H(f) (Theorem A.3), any single-bit mutation invalidates the confirmation token and aborts execution before network transmission.

核心防御逻辑：防御的是多智能体协作中的 TOCTOU（检查时与使用时不一致）漂移風險——防止人类刚审完预演计划，另一个后台 AI 就悄悄改了标题、标签或替换了视频文件。通过绑定元数据与 64KB 分块 SHA-256 视频哈希（定理 A.3），改一个字节就会立刻拦截。

Q6 [Reviewer #2 (Distributed Systems & State-Machine Formalism)]
EN: Browser DOM automation (Playwright) is notoriously fragile when platforms update their UI. How does your architecture handle DOM selector breakage?
CN: 网页端自动化（Playwright）在平台改版时很容易因为 DOM 选择器失效而崩溃，你们的架构是如何应对这种脆弱性的?

Standard SCI Rebuttal (English): Rather than assuming DOM selectors will never break, our architecture treats UI fragility as a first-class design constraint across three mechanisms: (1) Preflight & Dry-Run isolation checks selector availability and authentication validity before entering SUBMITTING; (2) explicit phase separation distinguishes pre-click failures (safe to abort into FAILED without side effects) from post-click verification failures (latched into UNKNOWN); and (3) a dedicated test suite of 2,737 unit and contract tests across 126 modules (Table 8, Supplementary Appendix F) continuously validates adapter contracts, fallback selectors, and receipt serialization.

核心防御逻辑：我们不假设网页永不改版，而是把“UI 脆弱性”作为一等公民来设计：提交前坏了直接安全退回 FAILED；点击提交后如果选择器变了导致读不到回执，状态机立刻锁入 UNKNOWN 防止重发；同时配合 126 个测试模块、2,737 个单元测试（表 8 与附录 F）持续验证适配器契约。

Q7 [Reviewer #3 (Empirical Evaluation & Statistical Validity)]
EN: Your empirical evaluation comes from a single organizational laboratory (Agentic Work Lab). How do you justify external validity and generalizability?
CN: 你们的实证数据来自单一机构实验室（Agentic Work Lab），如何证明研究结论的外部有效性（External Validity）与普适性?

Standard SCI Rebuttal (English): In Design Science Research and empirical software engineering (Hevner et al., 2004; Runeson & Höst, 2009; Wieringa, 2014), generalization from longitudinal single-site studies relies on 'analytic generalization' (architectural mechanisms and mathematical invariants) rather than purely statistical population sampling. While our operational dataset originates from one laboratory, the technical environment spans 3 distinct foundation model families (GPT/Codex, Claude, DeepSeek), 5 structurally diverse external platforms (Bilibili, YouTube, X, Douyin, Instagram) covering both REST APIs and browser-automated engines, 69 operational checkpoints, 55 published videos, 30 SHA-256 locked media versions (2.21 GB), and 2,737 automated tests. Furthermore, all schemas and source code are publicly archived under the MIT License so independent teams can replicate the governance gates in their own pipelines.

核心防御逻辑：依据软件工程实证案例研究规范（Runeson & Höst, 2009；Wieringa, 2014），设计科学采用的是“分析性推广（Analytic Generalization）”。虽然来自同一实验室，但横跨了 3 大主流模型家族（GPT、Claude、DeepSeek）、5 个异构国内外平台（API + 浏览器混合）、69 次运维事件、55 条视频与 2,737 个测试，且全部开源可复现。

Q8 [Reviewer #3 (Empirical Evaluation & Statistical Validity)]
EN: In Table 7, why do the AI-assisted cohorts ('Whale Girl Shift Log' mean = 95.89 views; 'GPT Duty Log' mean = 32.56 views) have lower views than the legacy baseline (mean = 3,477.00 views)? Doesn't this show AI performed worse?
CN: 在表 7 中，为什么 AI 辅助序列（鲸鱼娘均值 95.89，GPT 值班日志均值 32.56）的播放量反而低于历史老视频基线（均值 3,477）？这难道不是说明 AI 表现更差吗?

Standard SCI Rebuttal (English): Quite the opposite—highlighting this discrepancy is the core point of RQ3 and Algorithm 3 (Epistemic Audit Gate). As explained in Section 5.5, comparing raw cumulative views between legacy videos (which have accumulated organic traffic and algorithmic recommendations over many months/years) and newly launched AI-assisted series (published only days or weeks prior to snapshot collection) commits severe exposure-duration confounding. A naive or dishonest study would cherry-pick metrics or claim 'AI increased output volume by 300%'. Instead, our Mann–Whitney U test (U = 65.5, z = -3.24, p = 0.0012, r_rb = 0.721) formally quantifies this distributional shift and enforces our epistemic rule: operational governance success (0 unauthorized writes, 0 duplicate posts, 100% SHA-256 integrity) must be evaluated independently from time-confounded audience traffic.

核心防御逻辑：这恰恰是本文 RQ3 最亮眼的学术贡献（诚实审计门，算法 3）！老视频挂在网上累积了很久的自然流量，而新 AI 序列才刚发布不久。如果直接比播放量就犯了统计学上的“曝光周期混杂偏差（Exposure-Duration Confounding）”。我们专门用 Mann–Whitney U 检验（p=0.0012）揭示这一现象，证明工程治理的成功（0 越权、0 重复发布）绝不能与受时间干扰的播放量混为一谈——审稿人最喜欢这种极其诚实、反浮躁的严谨学风！

Q9 [Reviewer #3 (Empirical Evaluation & Statistical Validity)]
EN: In Table 3, 68 out of 69 operational events were blocked non-write checkpoints and only 1 was an executed write. Is N=1 write event too small to evaluate write execution?
CN: 在表 3 中，69 次事件里有 68 次是非写入拦截，只有 1 次是写入执行。1 次写入样本是不是太少了?

Standard SCI Rebuttal (English): Table 3 specifically measures the autonomous scheduled heartbeat daemon (DeepSeek / OpenClaw routine runner) governed by Algorithm 1 (planRun), whose primary safety requirement is to prevent unscheduled or duplicate background mutations across continuous polling intervals (68/68 = 100% suppression accuracy, 0 false-positive writes). The actual multi-platform media write and release executions are evaluated separately across the 5-platform release matrix (Table 4), the phase-level telemetry decomposition (Table 5), the 30 cryptographically verified video delivery versions totaling 2.21 GB (Table 6), the 55 published channel videos (Table 7), and 2,737 automated test cases simulating every state transition (Table 8).

核心防御逻辑：表 3 测的是后台定时巡检守护进程（Algorithm 1），它的核心使命就是在 68 个非写入时间窗口里 100% 拦住 AI 乱动文件；而真正的跨平台写入与发布实测在表 4（5 大平台发布）、表 5（Instagram 耗时拆解）、表 6（30 个视频版本 2.21GB 交付）、表 7（55 条已发布视频）和表 8（2,737 个自动化测试）中得到了全面验证。

Q10 [Reviewer #3 (Empirical Evaluation & Statistical Validity)]
EN: Why did you note that the 'blinded identical-task multi-model merchant comparison' is an unexecuted plan rather than running it before submission?
CN: 为什么你们在论文里诚实说明“多模型同题盲测对比”目前只是待执行计划（Unexecuted Plan），而不是编造或凑一组数据放进去?

Standard SCI Rebuttal (English): Rule 4 of our Epistemic Hygiene Protocol (Section 3.2) strictly mandates separating verified execution receipts from planned experimental designs. In our production workflow, GPT/Codex, Claude, and DeepSeek were assigned complementary functional roles (synthesis/QA, UI tooling, and scheduled monitoring, Table 10) rather than redundant identical tasks. Documenting the schema and priority protocol for future blinded comparisons while explicitly labeling its unexecuted status prevents selective reporting bias and upholds the highest standards of empirical software engineering transparency.

核心防御逻辑：在实际生产中，三个模型是按特长分工协作的（GPT 写脚本与验证、Claude 做界面、DeepSeek 做定时巡检，见表 10），而不是让三个模型重复干同一件事。在论文中明确把“同题盲测”标注为待执行计划，体现了第 3.2 节提出的“诚实学术边界准则（Epistemic Hygiene）”。

Q11 [Reviewer #2 (Systems Performance & Overhead)]
EN: Does computing 64-KB chunked SHA-256 hashes across gigabytes of video files introduce unacceptable latency overhead into the release pipeline?
CN: 对几个 GB 的视频文件做 64-KB 分块 SHA-256 加密哈希计算，会不会给发布流水线带来很大的性能延迟开销?

Standard SCI Rebuttal (English): No. Stream-based 64-KB SHA-256 hashing in Python/OpenSSL processes over 500–800 MB/s on modern NVMe SSDs, requiring less than 0.15 seconds per 75 MB video file while keeping peak RAM usage under 64 KB. Compared to external platform upload and confirmation latencies (e.g., 412.217 s on Instagram, Table 5), cryptographic hashing overhead represents less than 0.04% of end-to-end release time while guaranteeing 100% byte-level payload immutability.

核心防御逻辑：64-KB 流式 SHA-256 哈希在现代固态硬盘上每秒可处理 500–800 MB，算一个 75 MB 的视频不到 0.15 秒，内存占用仅 64 KB；相比于跨平台上传动辄 412 秒的耗时（表 5），哈希校验的开销占比不到 0.04%，却换来了 100% 的字节级防篡改安全。

Q12 [Reviewer #1 (Human Factors & Cognitive Load)]
EN: Does routing UNKNOWN and NEEDS_HUMAN states to the human Project Manager create alert fatigue or bottleneck scalability?
CN: 把 UNKNOWN 和 NEEDS_HUMAN 状态交给人类项目经理处理，会不会导致人类“告警疲劳（Alert Fatigue）”或成为扩展性瓶颈?

Standard SCI Rebuttal (English): Our architecture minimizes human cognitive load by filtering out all routine checks and pre-submission errors automatically: Layer 1 suppressed 68/69 non-write polling intervals without human involvement, and Layer 3 only escalates to NEEDS_HUMAN or UNKNOWN when an irreversible external boundary (such as an expired login session or post-submit read-back timeout) occurs. Moreover, when escalation occurs, the receipt provides exact diagnostic telemetry (platform, phase latency, error code, and pre-filled CLI recovery commands), reducing human resolution to a single verification action rather than manual log debugging.

核心防御逻辑：恰恰相反，第一层门控自动过滤了 68/69 的日常巡检，只有遇到真正不可逆的外部边界（如登

录过期或提交后超时）才会触发人工接管，而且回执单里直接给出了精确的故障阶段与恢复命令，极大降低了人类的排查负担。

Q13 [Reviewer #2 (State Machine Completeness)]
EN: In Figure 3 and Algorithm 2, are the eight states of the release receipt state machine dead-lock free and formally terminating?
CN: 在图 3 和算法 2 中，发布回执状态机的 8 个状态是否保证无死锁（Deadlock-Free）且必然终止?

Standard SCI Rebuttal (English): Yes. As formalized in Section 4.3 and proven in Supplementary Appendix A (Theorem A.2), the state transition graph is a Directed Acyclic Graph (DAG) within any single execution attempt: every active state (INIT, DRY_RUN, PREFLIGHT, SUBMITTING, VERIFYING) has bounded timeout guards (tau_submit, tau_verify) that deterministically transition into one of the four absorbing terminal states (PUBLISHED, UNKNOWN, NEEDS_HUMAN, FAILED). Consequently, no execution path can loop infinitely or deadlock.

核心防御逻辑：在单次执行中，状态转移图是一个带超时上界（tau_submit, tau_verify）的有向无环图（DAG），任何活跃状态在有限时间内必然进入四个吸收态（PUBLISHED、UNKNOWN、NEEDS_HUMAN、FAILED）之一，因此在数学上绝不存在死锁或无限循环。

Q14 [Reviewer #3 (Ablation Study Design)]
EN: How were the ablation configurations (A1–A5 in Table 11) evaluated without intentionally corrupting live public social media accounts?
CN: 表 11 中的消融实验（A1–A5）是如何在不污染真实公开社交媒体账号的前提下完成验证的?

Standard SCI Rebuttal (English): As detailed in Section 5.9 and Table 11, each ablation variant (A1–A5) was verified by combining historical incident traces observed prior to gate deployment (e.g., the Douyin read-back timeout trace and Instagram session-expiry trace) with deterministic fault-injection test harnesses in the 2,737-test suite (e.g., tests/test_state_machine_*.py, tests/test_preflight_*.py, and tests/plan-run.test.mjs). This allows exact measurement of failure behavior under gate removal without spamming live public platforms.

核心防御逻辑：结合了门控上线前的真实故障历史记录（如抖音回读超时、Instagram 登录过期记录）以及 2,737 个自动化测试中的确定性故障注入测试（Fault-Injection Test Harness），既精确验证了去掉每一层门控后的必现故障，又避免了往真实公网平台恶意连发垃圾视频。

Q15 [Editor-in-Chief (Journal Fit & Readership Value)]
EN: What concrete takeaway can software engineering and project management practitioners immediately apply from your paper?
CN: 软件工程与项目管理的一线从业者能从你的论文中立刻拿走什么可落地的实践经验?

Standard SCI Rebuttal (English): Practitioners gain three immediately actionable engineering patterns: (1) Never rely on conversational prompts to constrain multi-agent write authority—enforce a pure scheduling gate and cryptographic plan-hash token (--confirm CURRENT_PLAN_HASH); (2) Never allow an AI agent to auto-retry a failed post-submission check—latch post-submit timeouts into an UNKNOWN state and reconcile via read-only queries; and (3) Always apply non-parametric cohort stratification before claiming AI productivity or traffic gains. All patterns are available as ready-to-run open-source tools in Agentic Work Lab and PublishOps CrossCast.

核心防御逻辑：给业界提供了三条即插即用的工程铁律：① 绝不用提示词约束 AI 写权限，必须用纯函数门控

> **+加密计划哈希锁；② 绝不允许 AI 在提交后超时自动重试，必须锁入 UNKNOWN 态做只读回查；③ 评估 AI 效果必须做非参数批次分层检验。且全套工具已开源。**